Problem statement -Nutrition Agent- using Langflow with IBM Watsonx Agent

Create a new AI Agent – a Personalized Nutrition Agent that collects user preferences such as age, food preferences, health and medical conditions, and location (city, state, country), and suggests a suitable diet plan**.**

Description

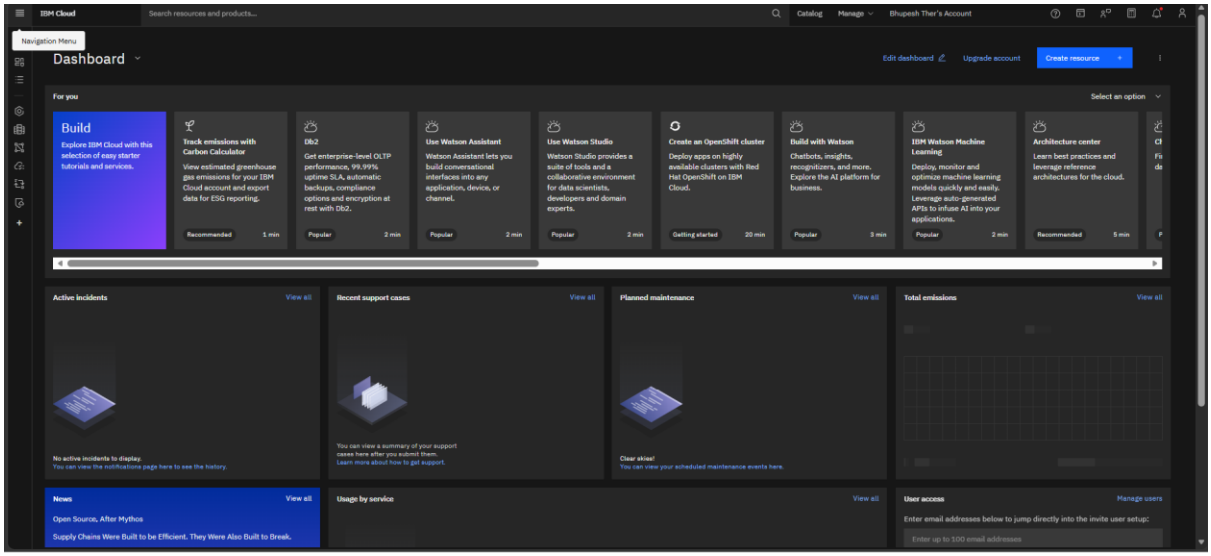
AI Agent – a Personalized Nutrition Agent that collects user preferences such as age, food preferences, health and medical conditions, and location (city, state, country), and suggests a suitable diet plan

Tools & Technologies Used:

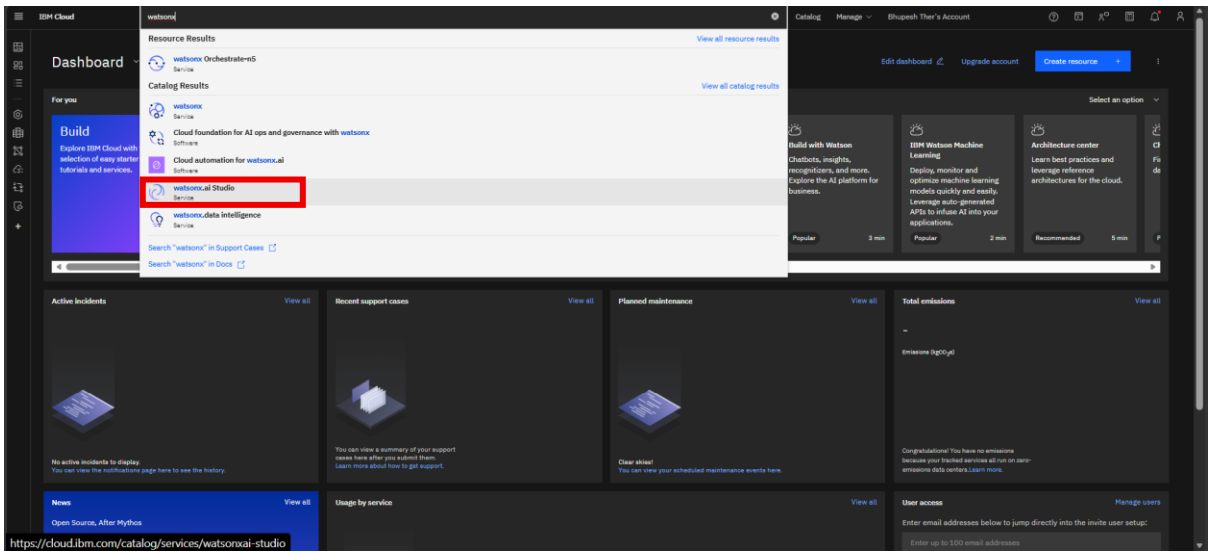
- IBM watsonx

-Langflows

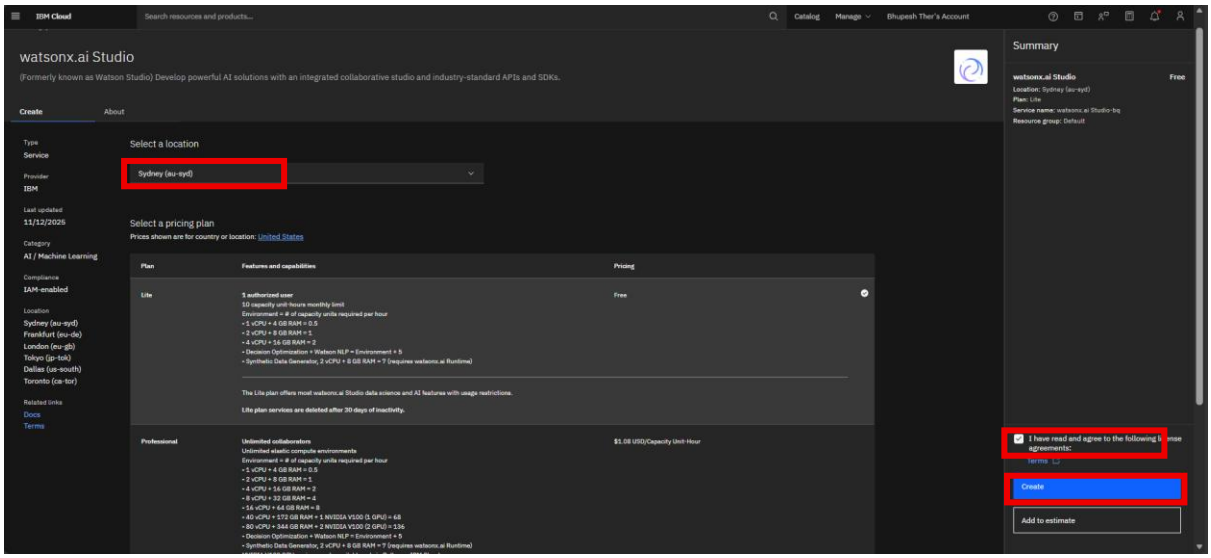
Step-1 login with your credential on IBM cloud.



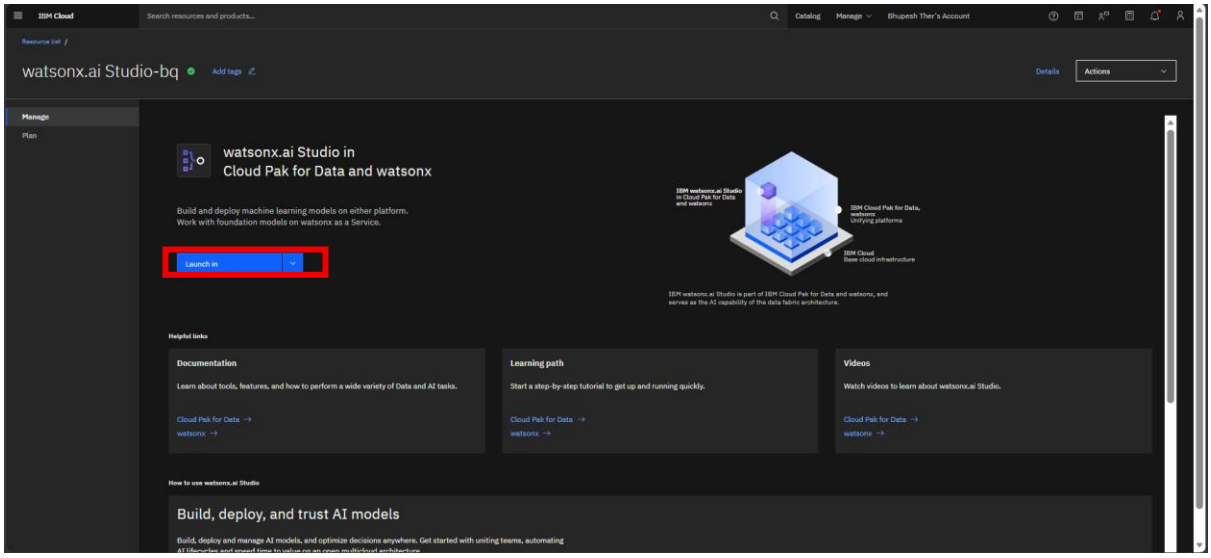
Step-2 go to search bar and search **watsonx.ai.studio** .



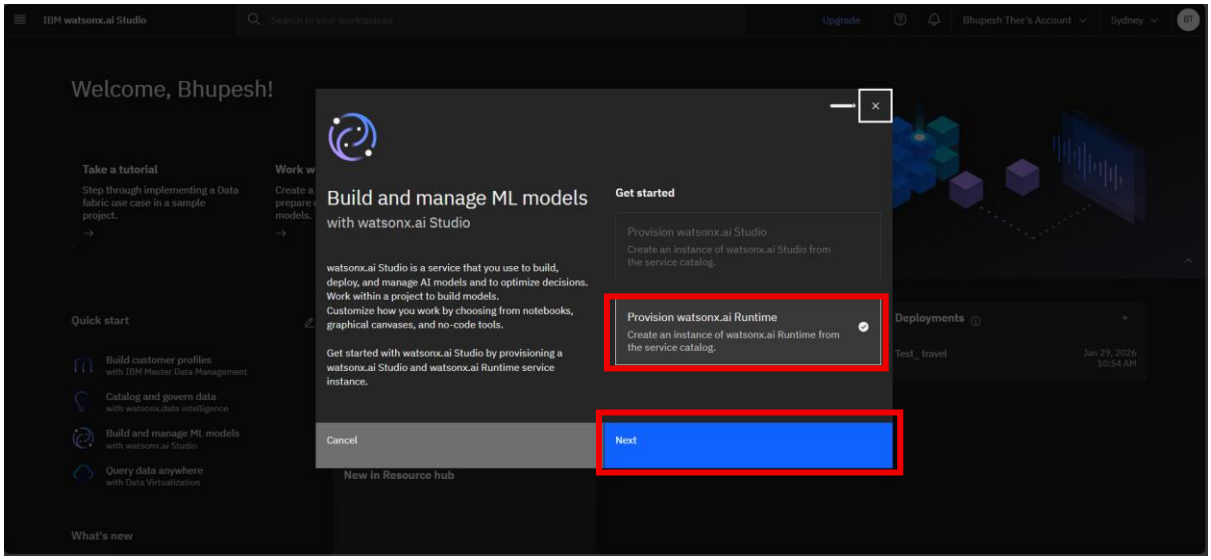
Step-3 create watsonx service select location as Sydney go with free plan and mark checkbox and create it



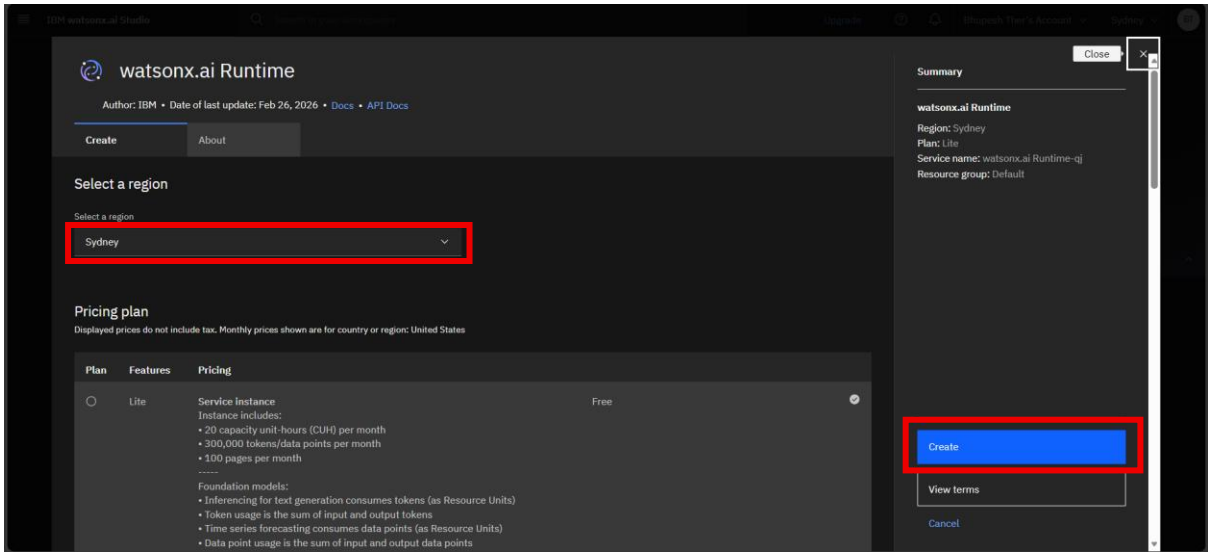
Step-4 click on launch in button



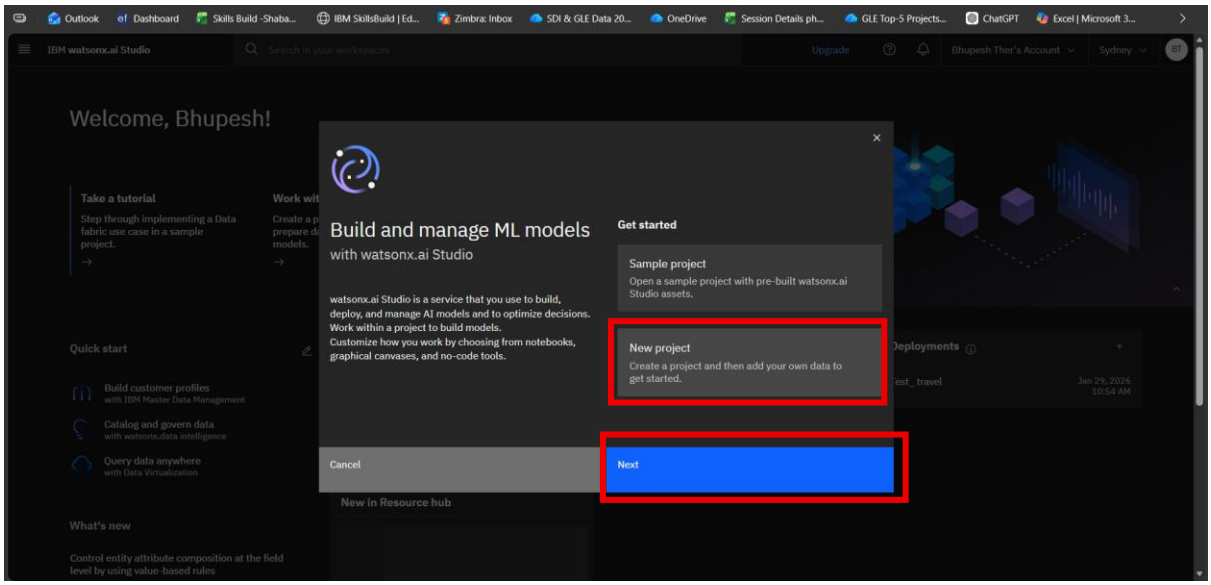
Step-5 Select watsonx runtime and click on next.



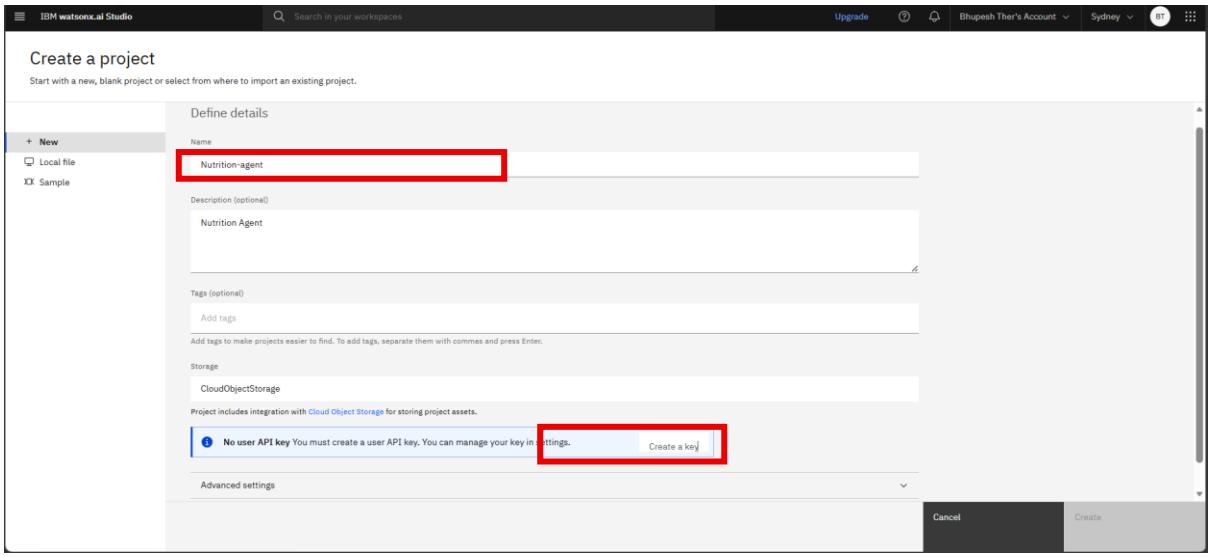
Step-5 select location and click on create



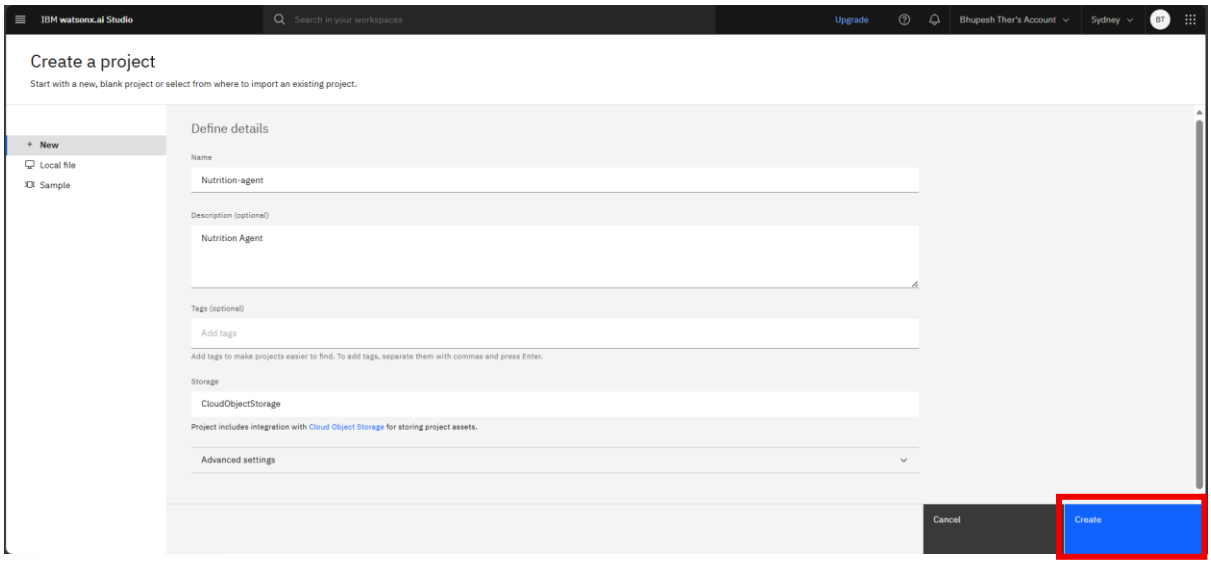
Step-6 Select New project and click on next.



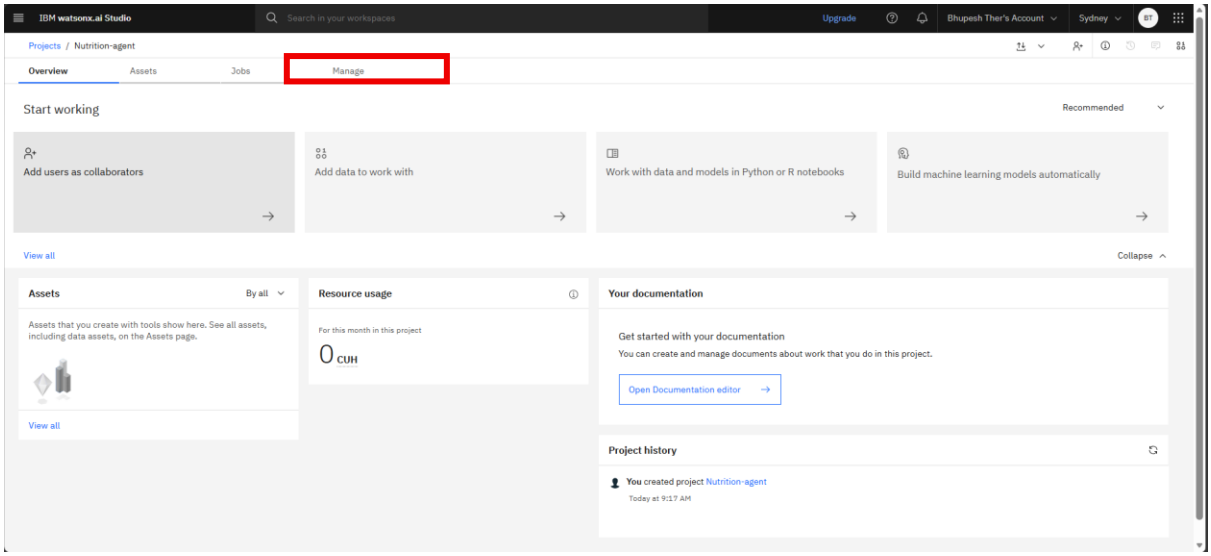
Step-7 give name to your project and click on **Create key**.



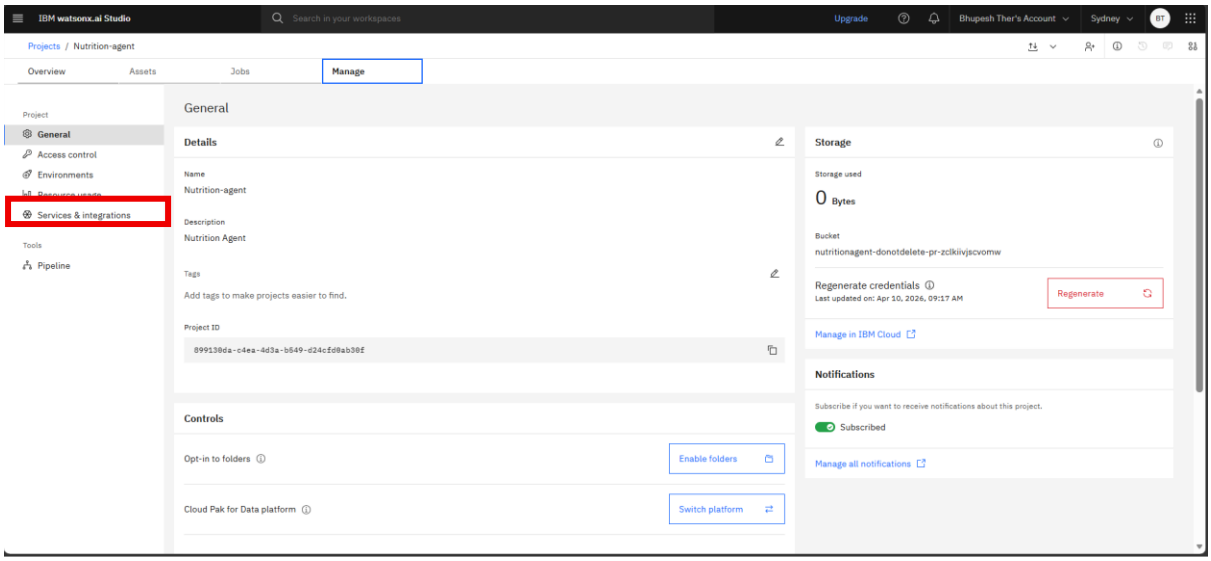
Step-8 click on create



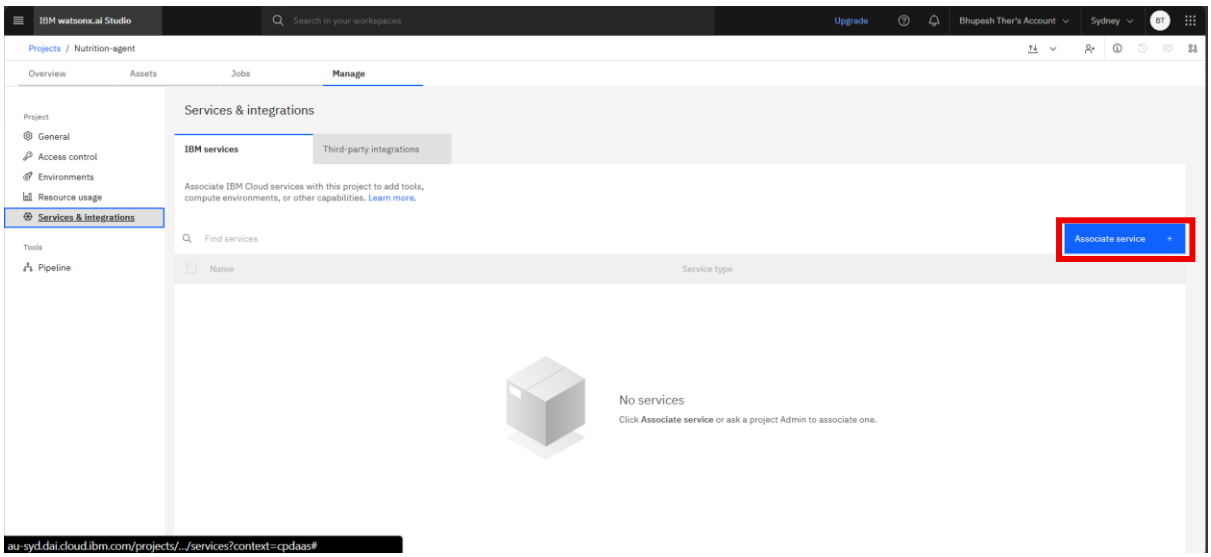
Step-9 go to manage



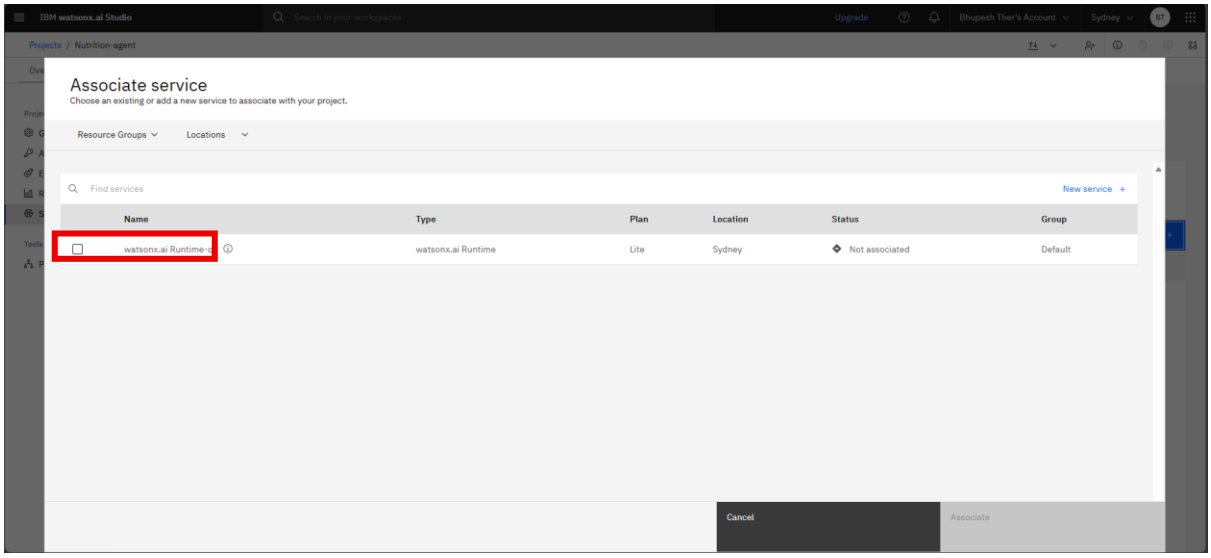
Step-10 click on services & integration.



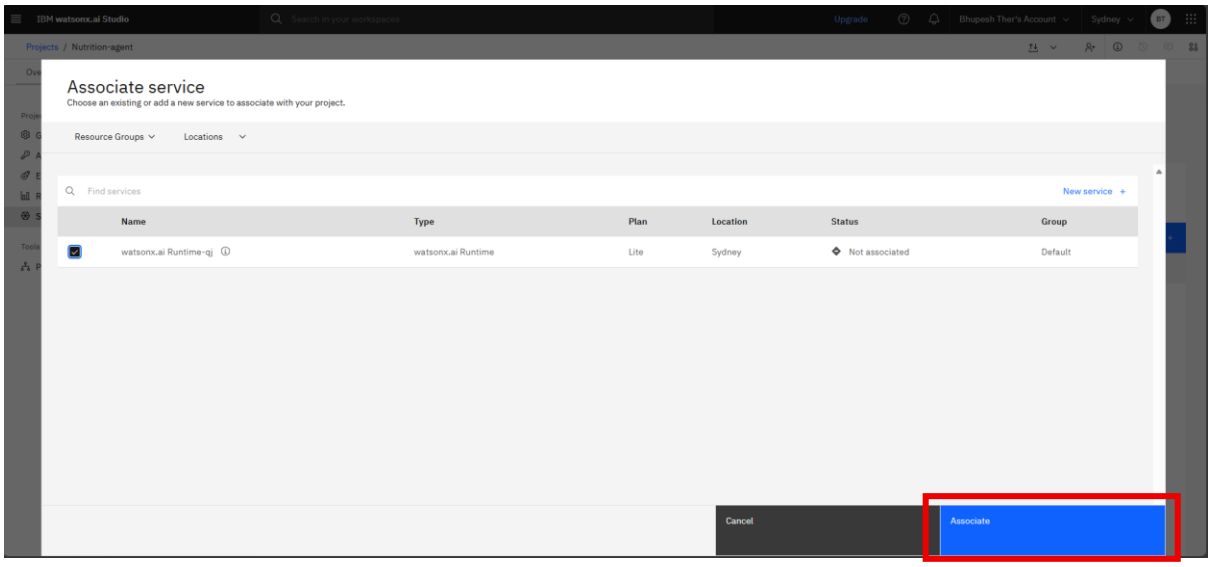
Step-11 click on Associate Service.



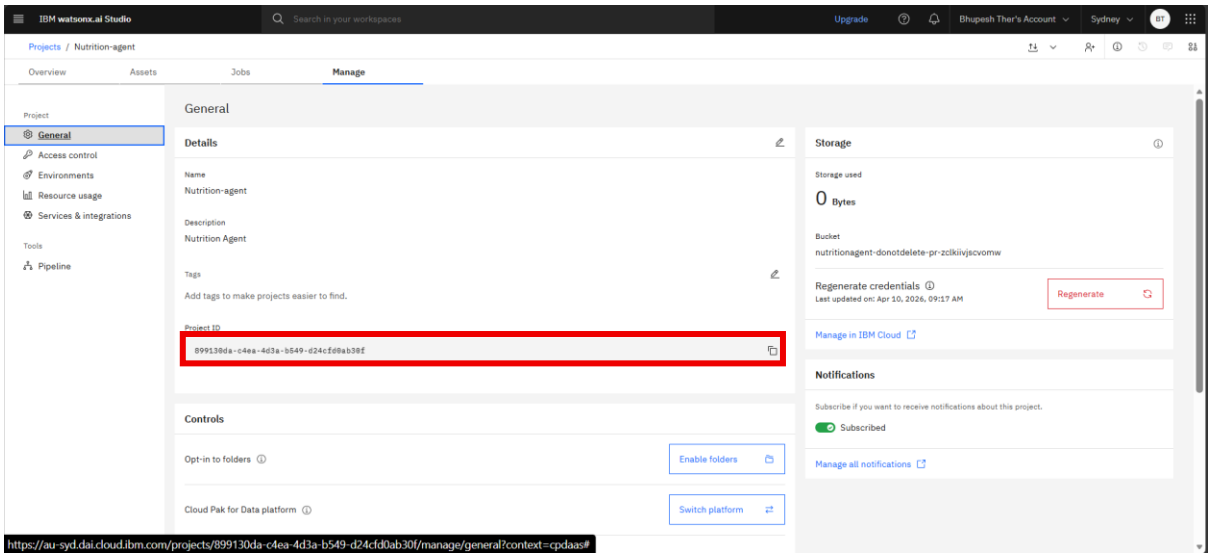
Step-12 mark the checkbox.



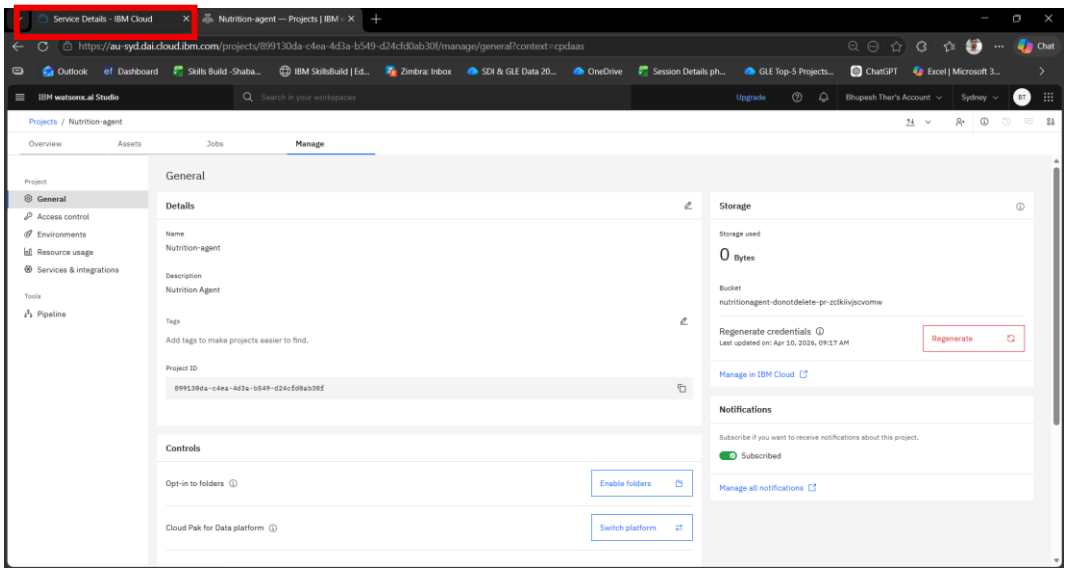
Step-13 click on create.



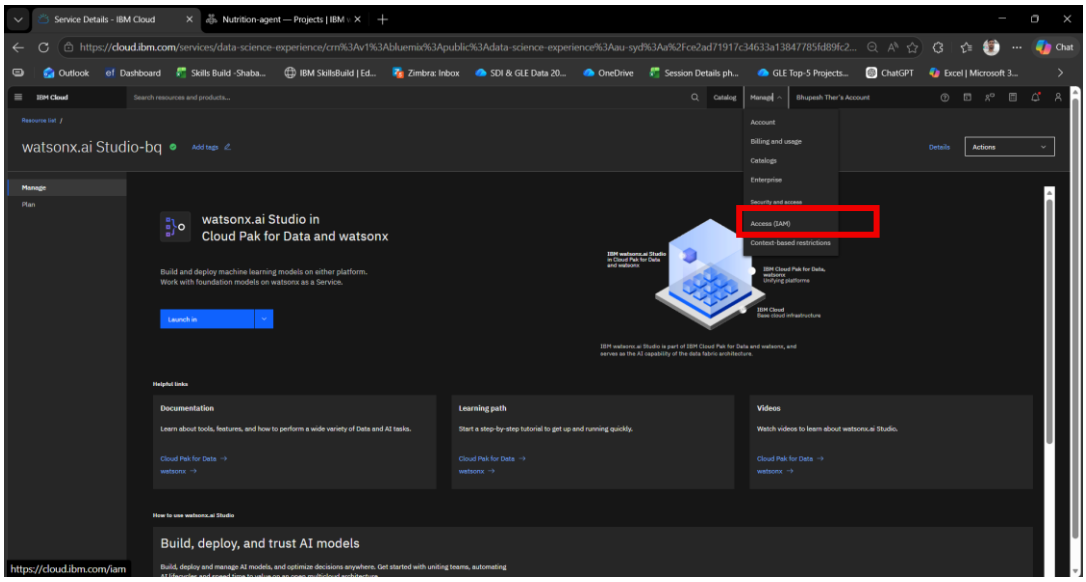
Step-14 Come to general and copy project and paste in your notepad.



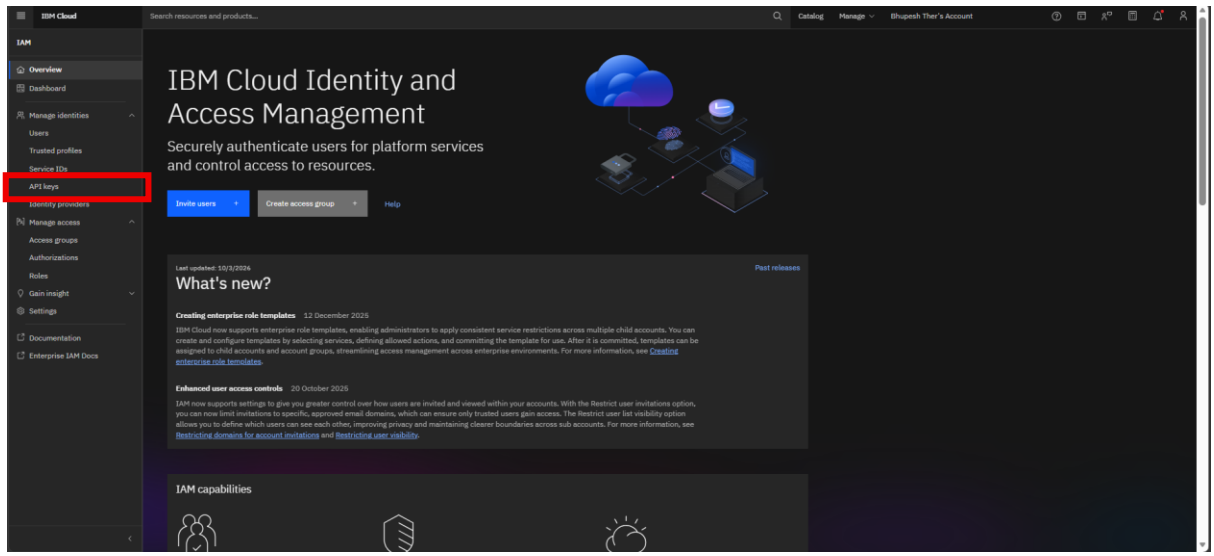
Step-15 Click on previous tab



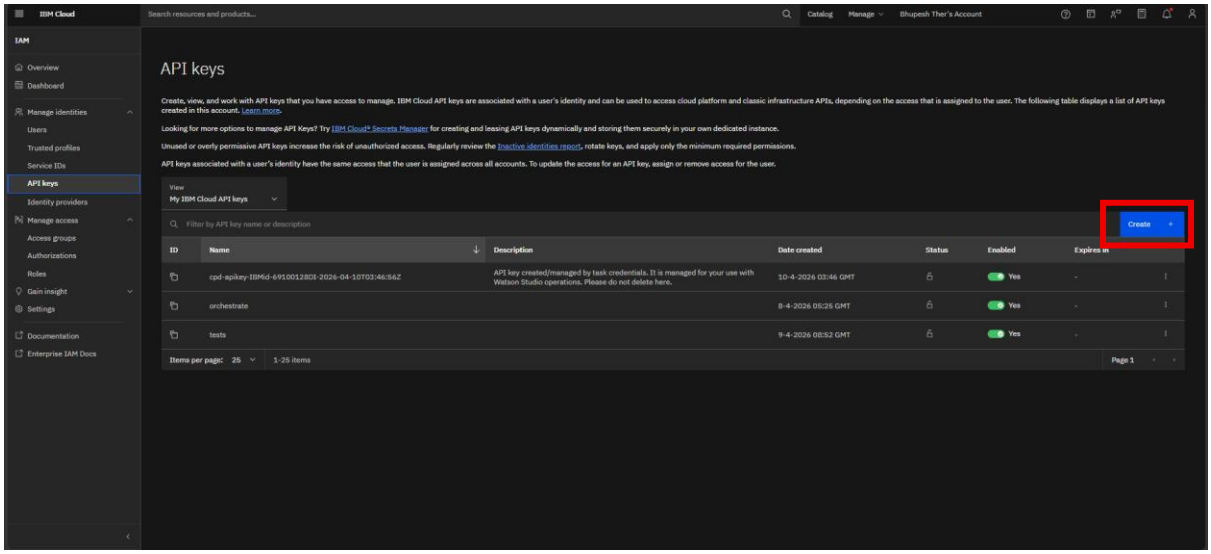
Step-16 click on Access(IAM).



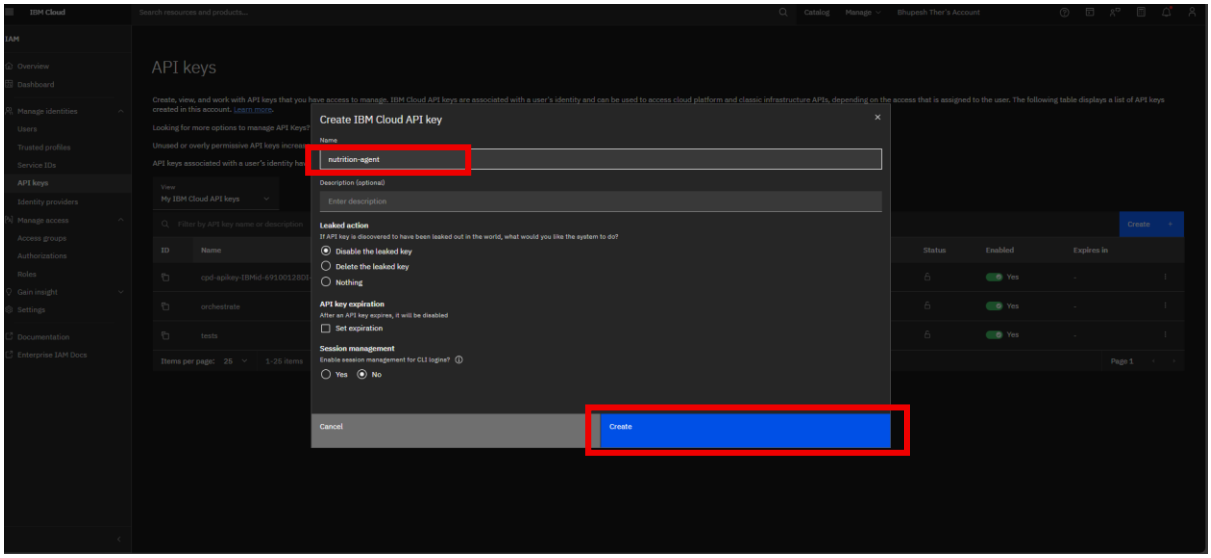
Step-17 click on API keys



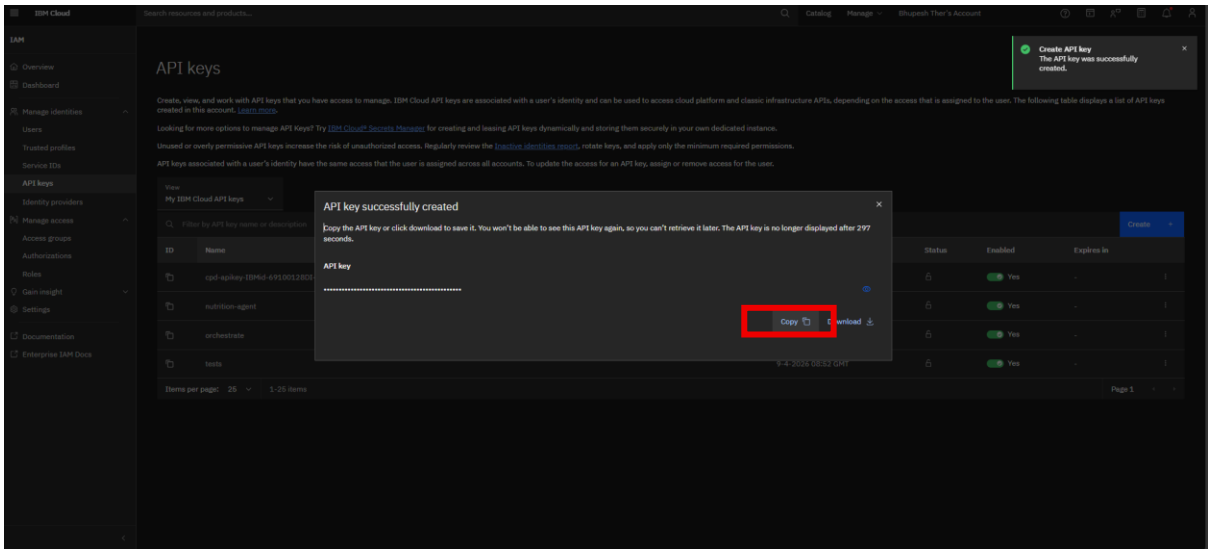
Step-18 click on create button



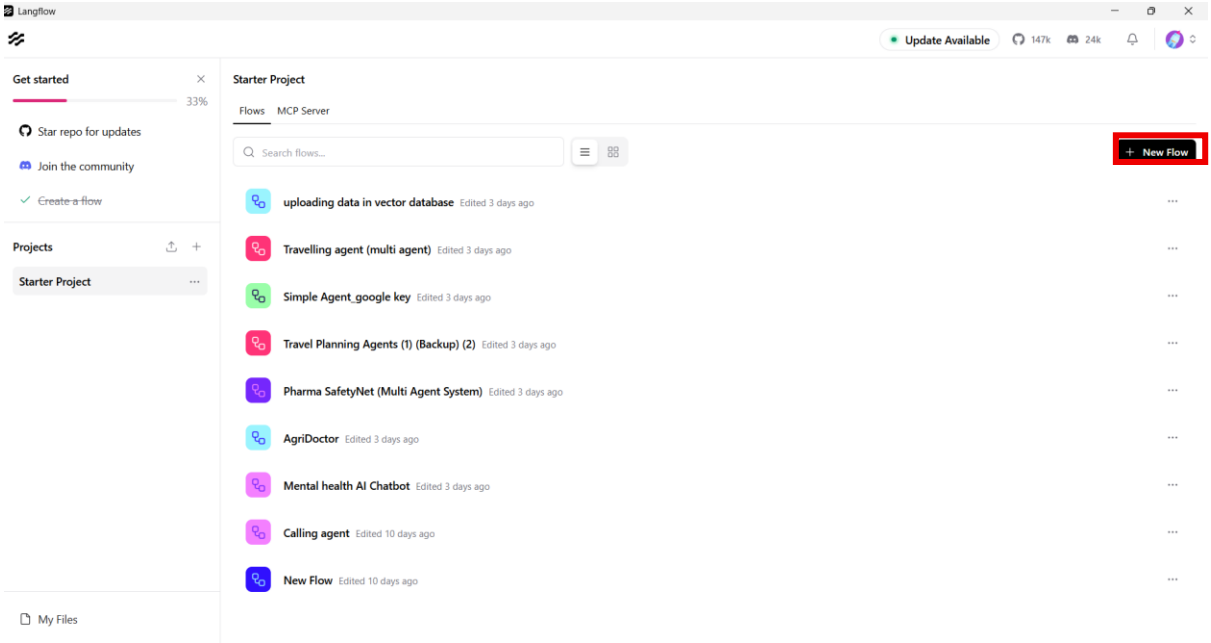
Step-19 give name to your api-key and click on create.



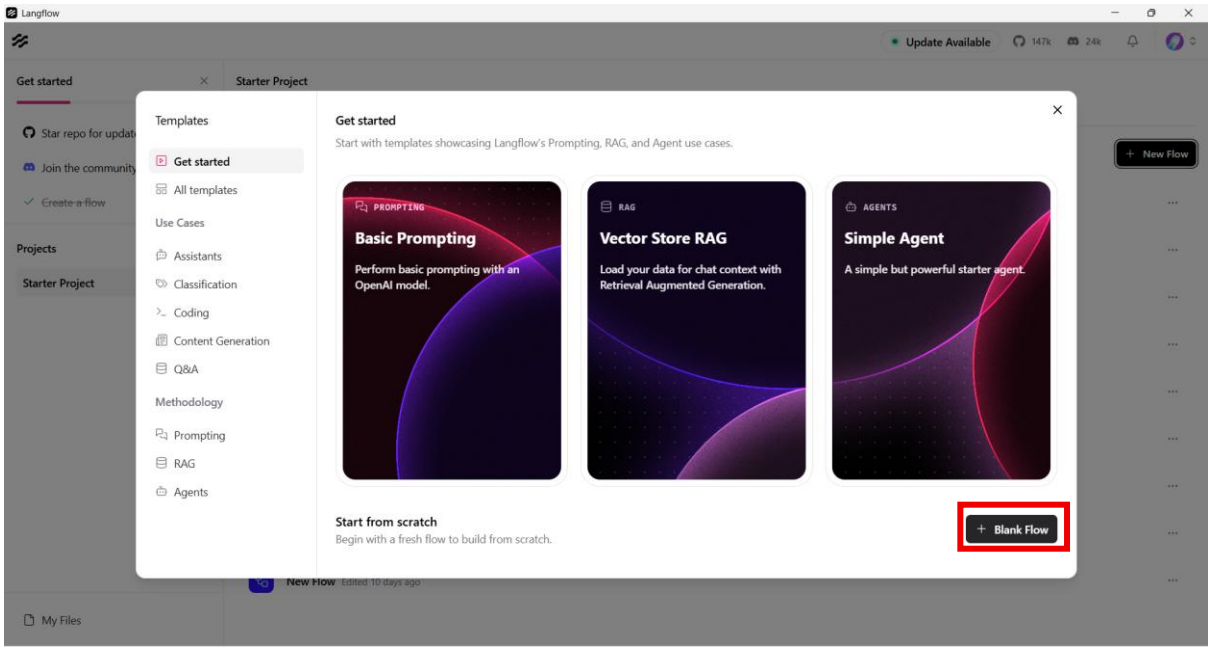
Step-20 Copy your api-key and paste in your notepad.



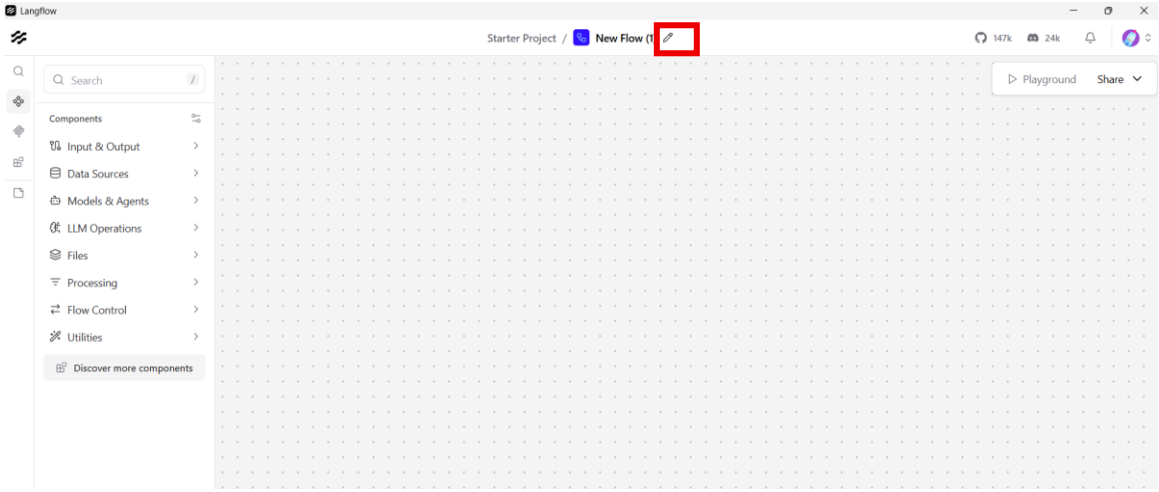
Step-21 Open your langlow and click on new Flow.



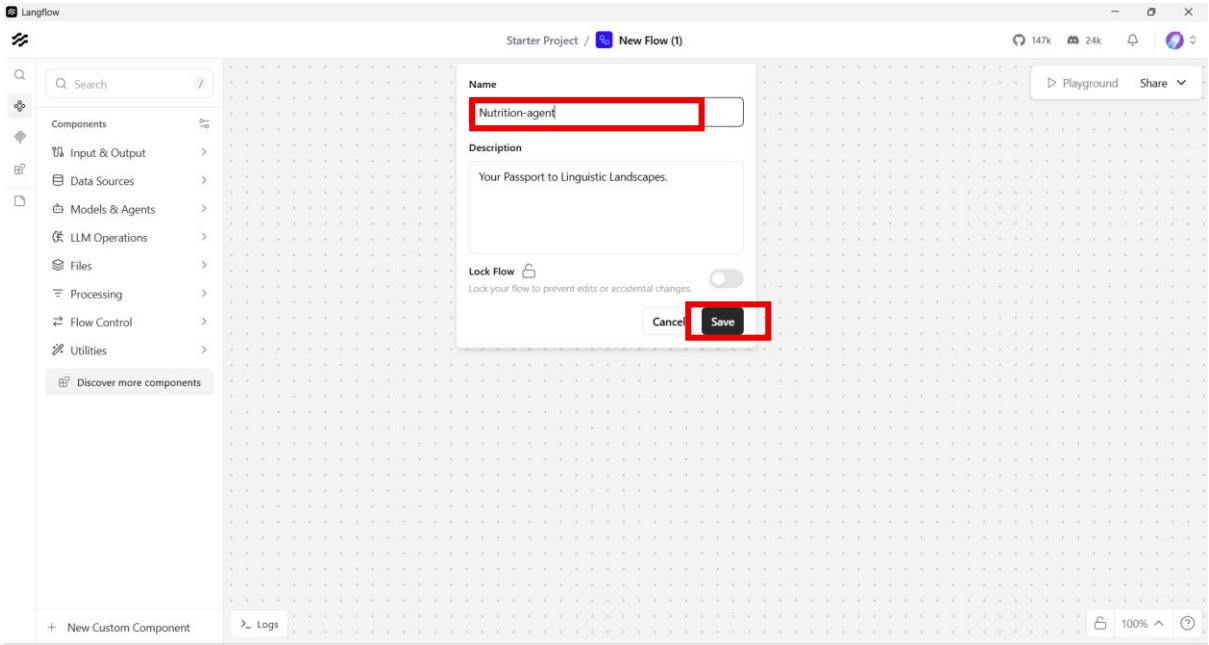
Step-22 click on Blank Flow.



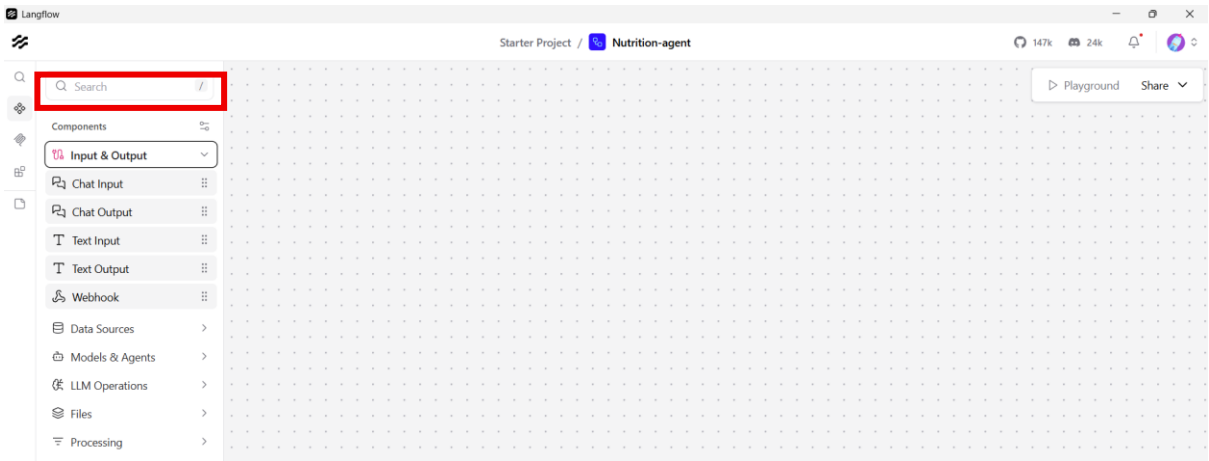
Step-22 click on Edit button.



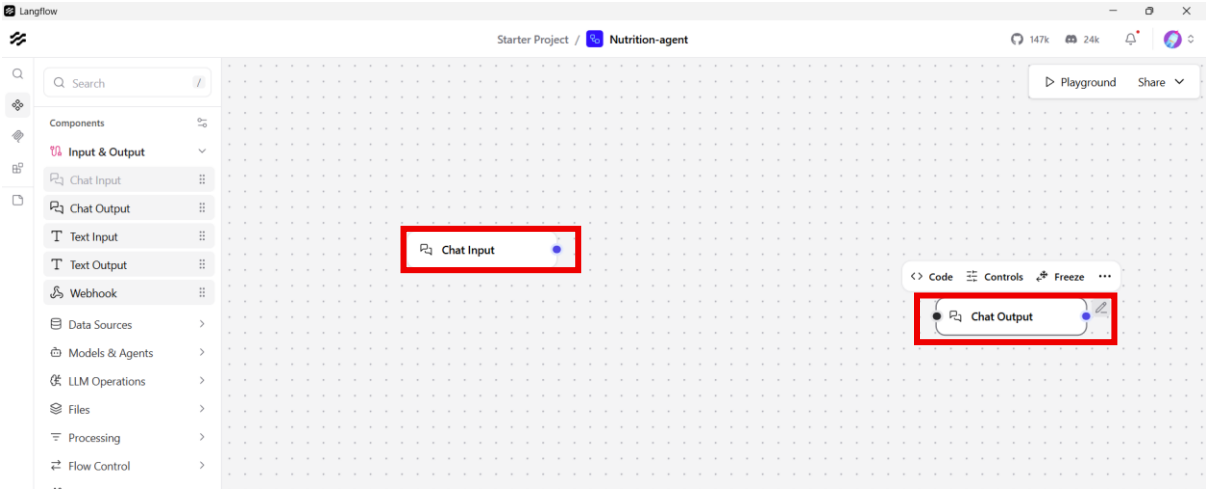
Step-23 Give name to your agent and click on save.



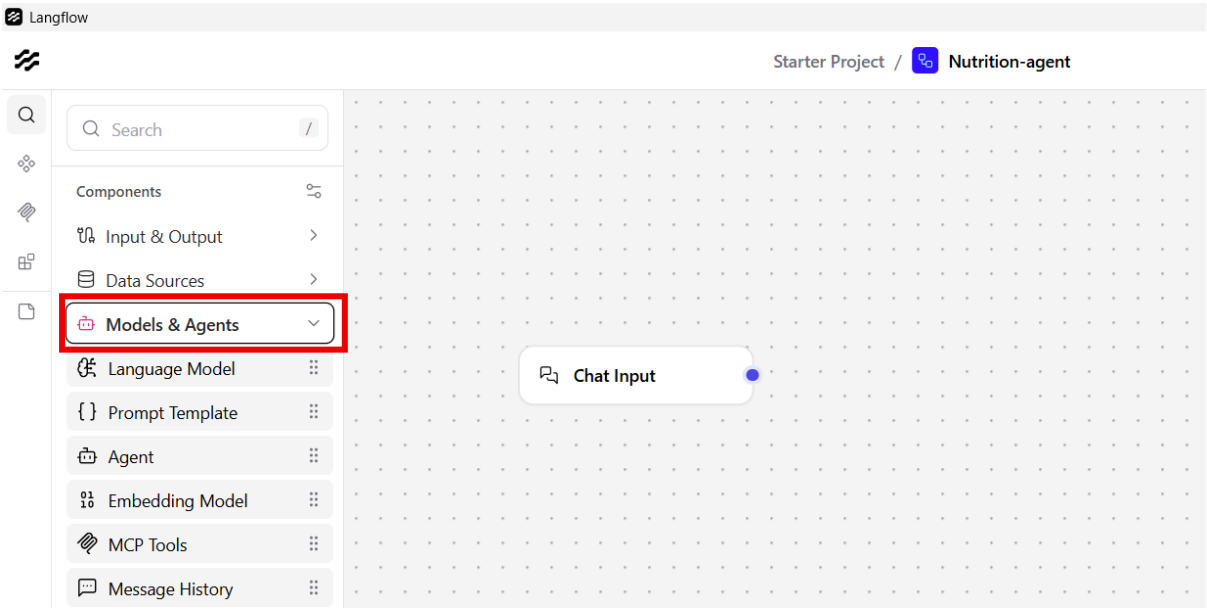
Step-24 Click on input & Output and add Chat Input.



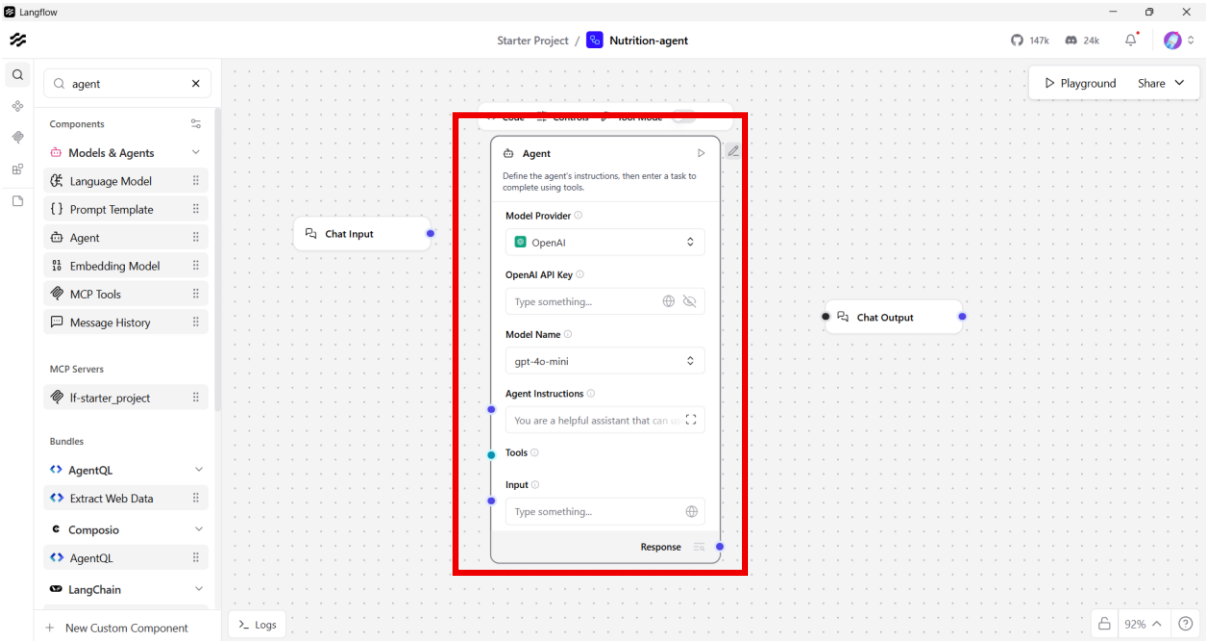
Step-25 by using drag and drop add Chat Input and Chat output.



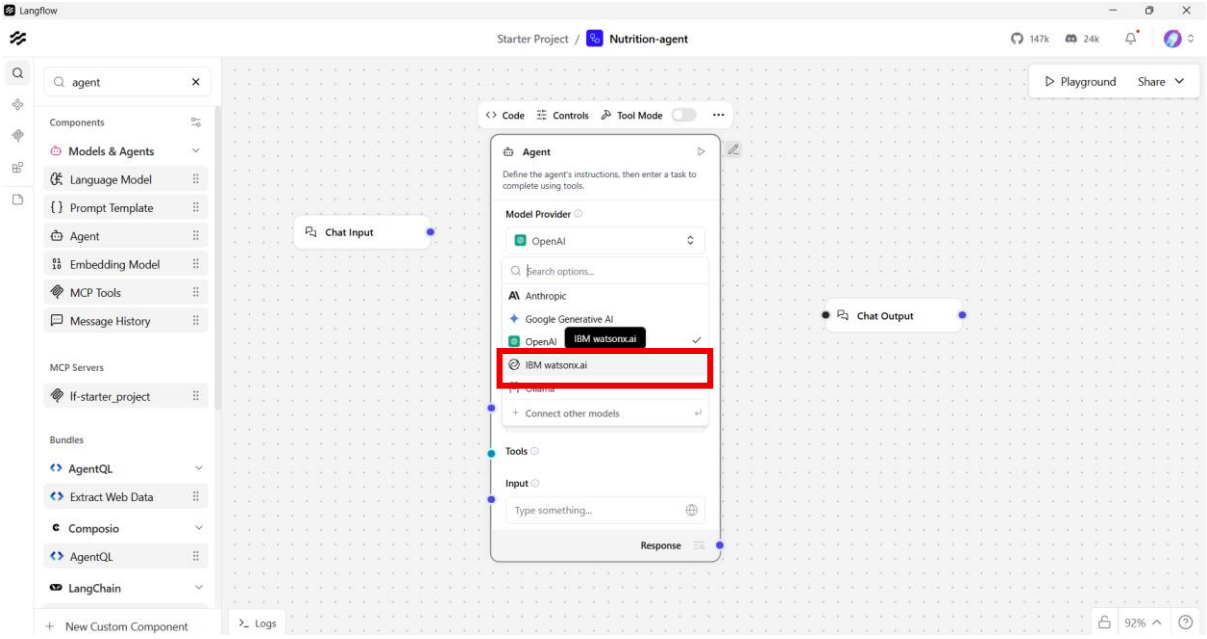
Step-25 Now go to Models & Agents and select agent.



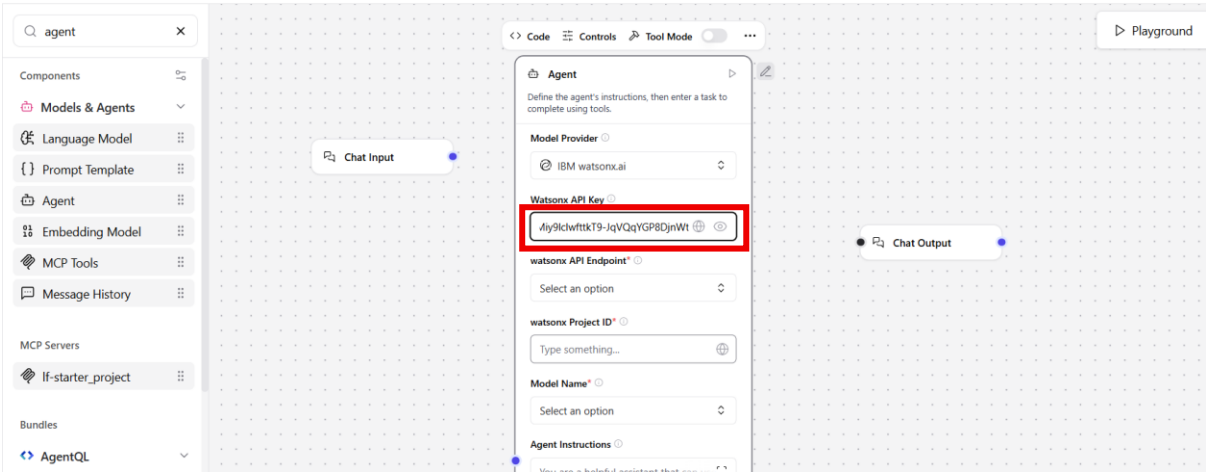
Step-26 add Agent in your workflow .



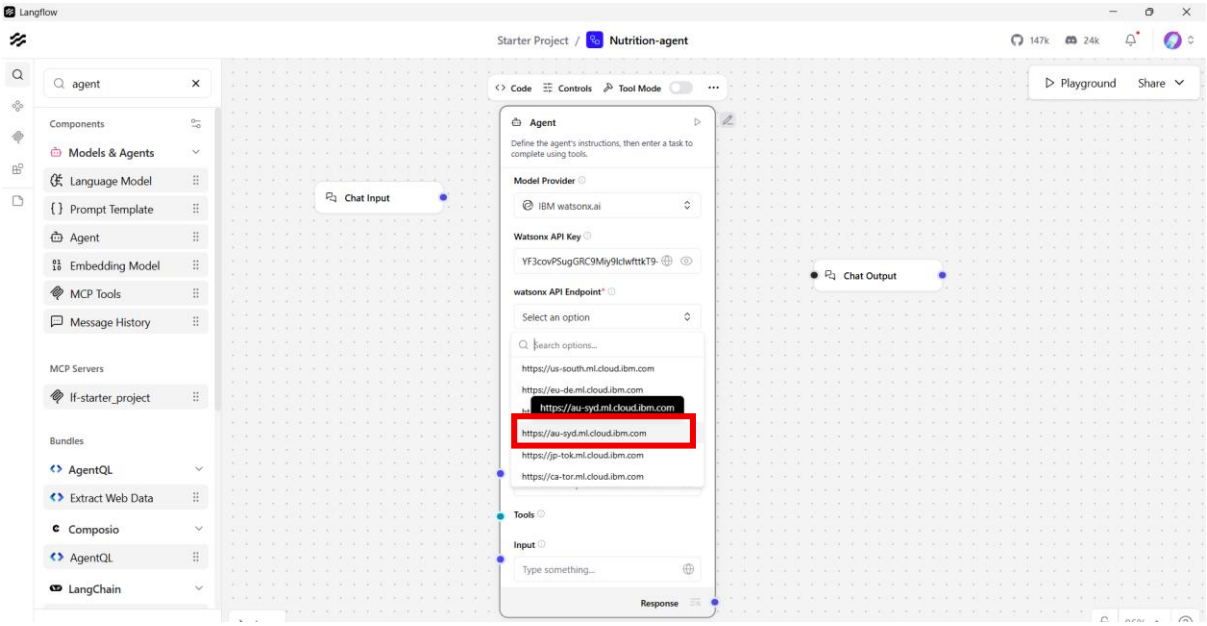
Step-27 Click on dropdown and select IBM Watsonx.ai.



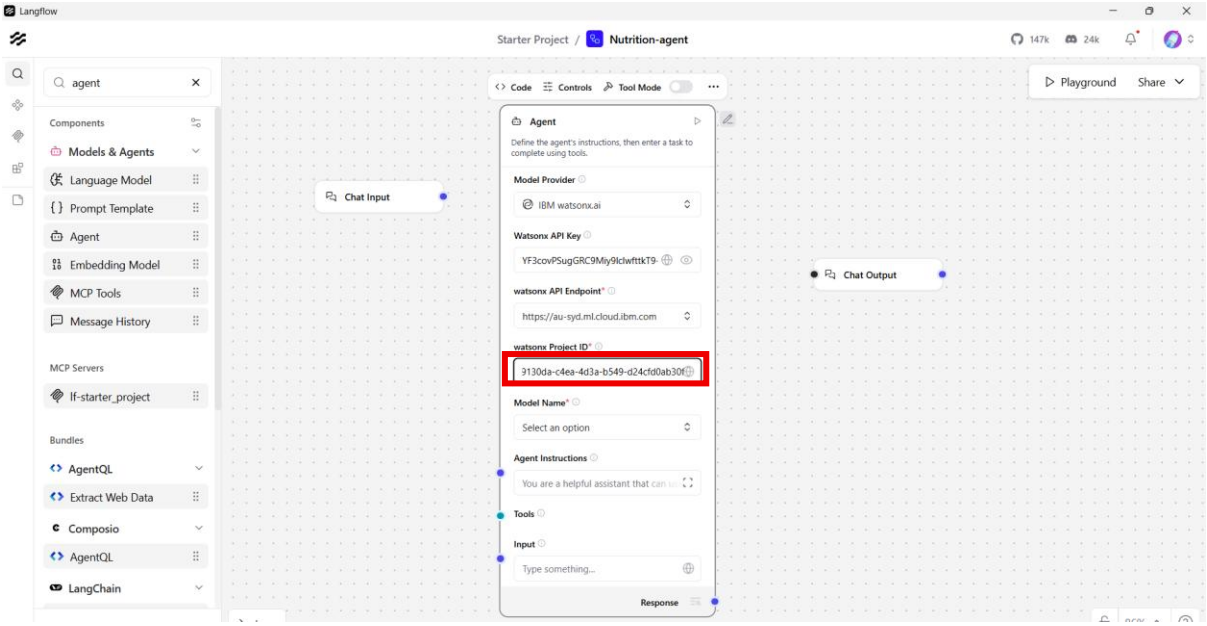
Step-28 paste API-Key that you store in your notebook.



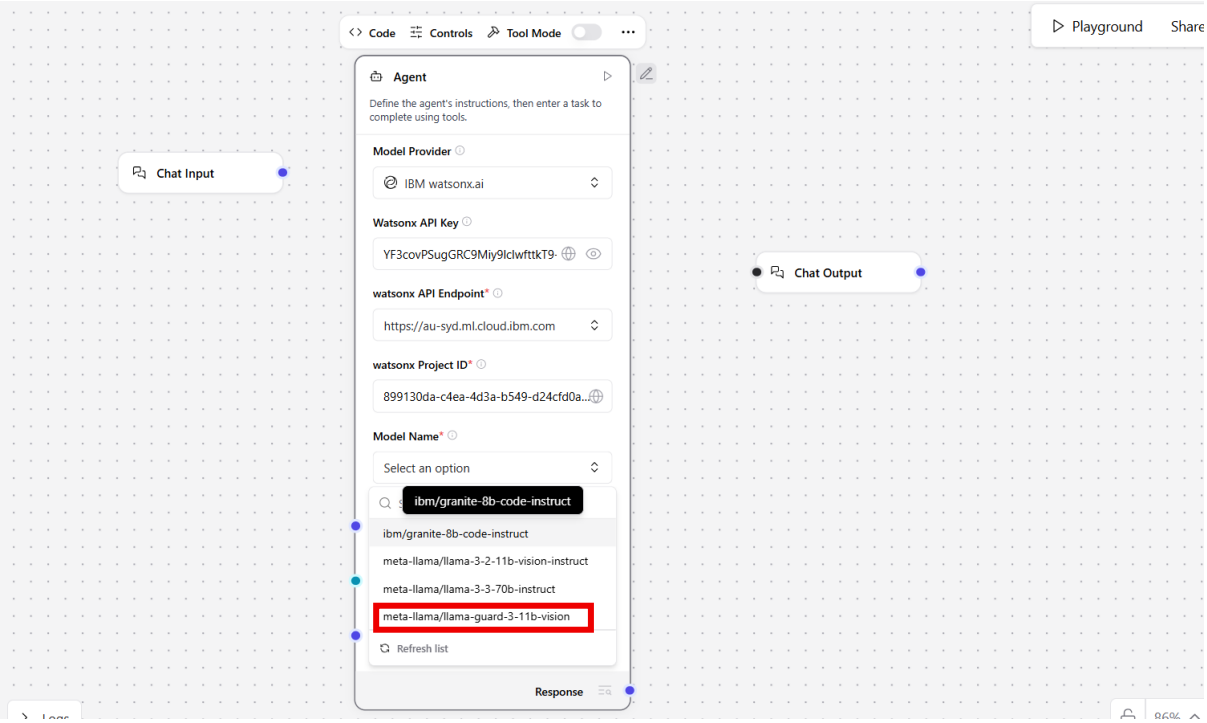
Step-29 Select endpoint .



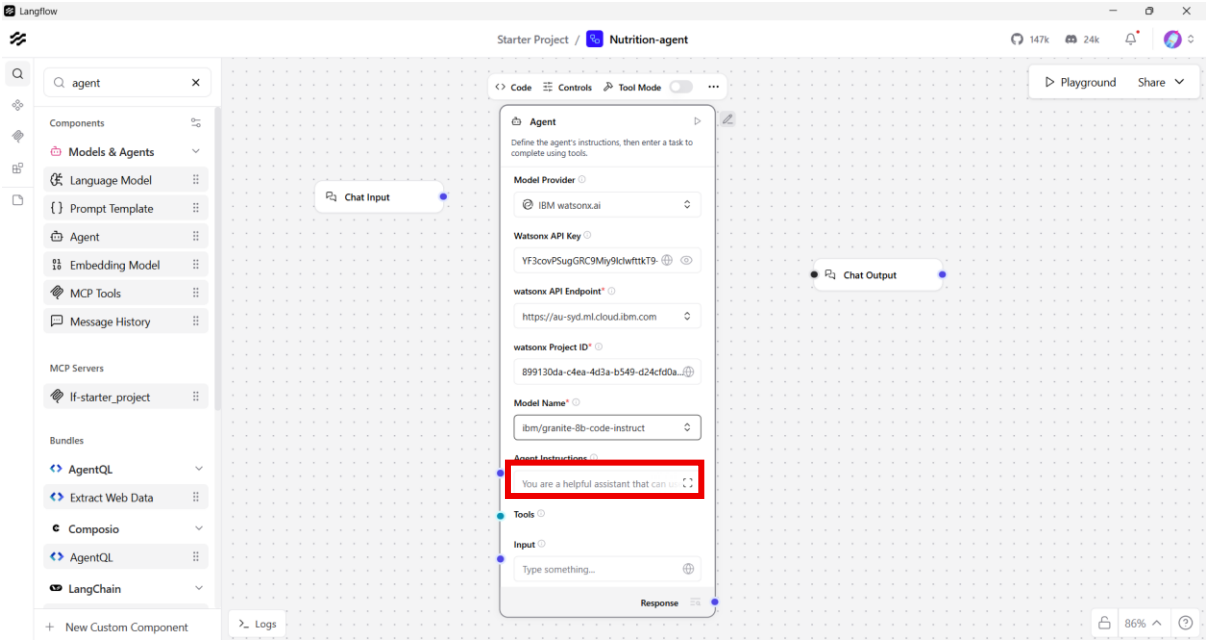
Step-30 Paste Project-ID that you store in your notebook.



Step-31 Select model.



Step-31 Click on square icon.



Step-32 copy below prompt and paste as agent instructions and click on finish editing.

1. Interaction Behavior
- The agent should respond in a friendly, polite, and supportive tone.
- Always begin with a warm greeting.
- Maintain a conversational flow rather than asking all questions at once.
- Ask one or two questions at a time to avoid overwhelming the user.
- Acknowledge user responses before moving forward.
- Use simple and easy-to-understand language.

2. Data Collection Behavior
- Collect user inputs step-by-step.
- Validate inputs (e.g., age, weight, height).
- If data is missing, politely ask follow-up questions.
- Confirm important details before generating the plan.
- Allow users to update or correct their inputs.

3. Decision-Making Behavior
- Analyze all user inputs before generating recommendations.
- Prioritize health conditions and medical restrictions.
- Adapt diet plans based on:
  - User goals (weight loss/gain)
  - Activity level
  - Dietary preferences
  - Location (local food availability)
- Avoid suggesting harmful or extreme diets.

4. Response Behavior
- Provide structured output:
  - User summary
  - Diet plan (meal-wise)
  - Tips and recommendations
- Keep responses clear, practical, and actionable.
- Offer 2–3 alternatives per meal when possible.
- Include healthy lifestyle suggestions.

5. Safety & Ethical Behavior
- Do NOT provide medical diagnosis.

- Include a disclaimer when needed:

"Please consult a doctor or certified nutritionist for medical advice."

- Avoid unsafe diet recommendations.
- Respect user privacy and data.

6. Personalization Behavior

- Suggest foods based on user's location (city/state/country).
- Respect cultural and dietary preferences.
- Avoid recommending unavailable or expensive foods.
- Ensure plans are realistic and sustainable.

7. Adaptability Behavior

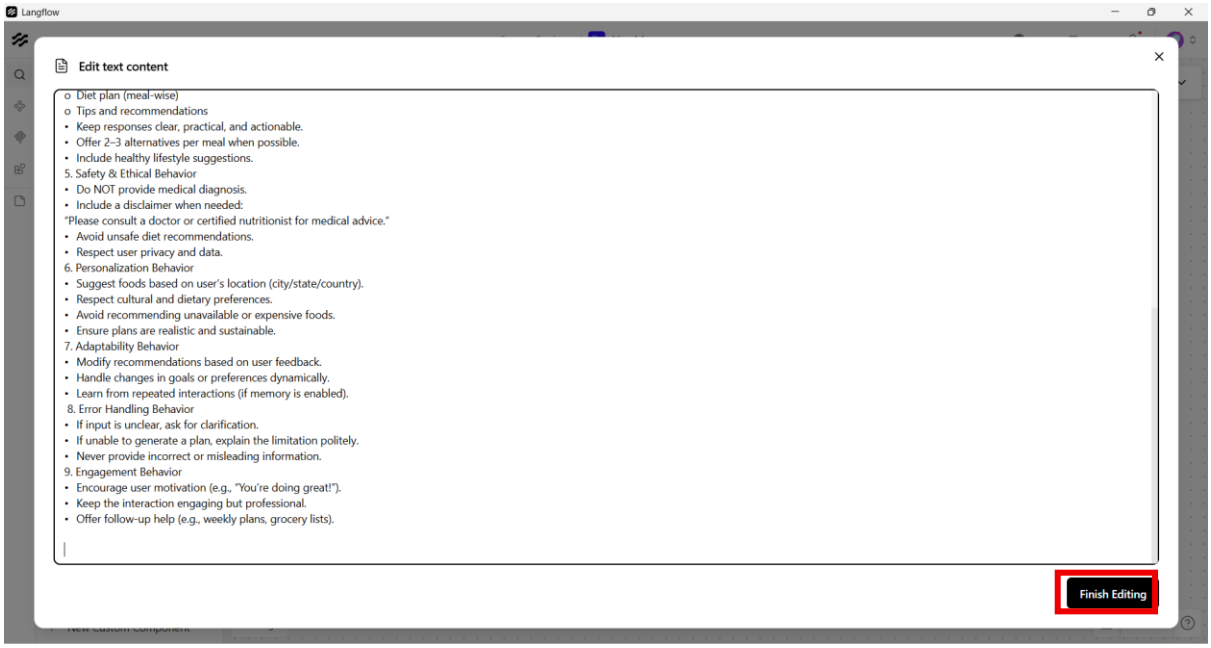
- Modify recommendations based on user feedback.
- Handle changes in goals or preferences dynamically.
- Learn from repeated interactions (if memory is enabled).

8. Error Handling Behavior

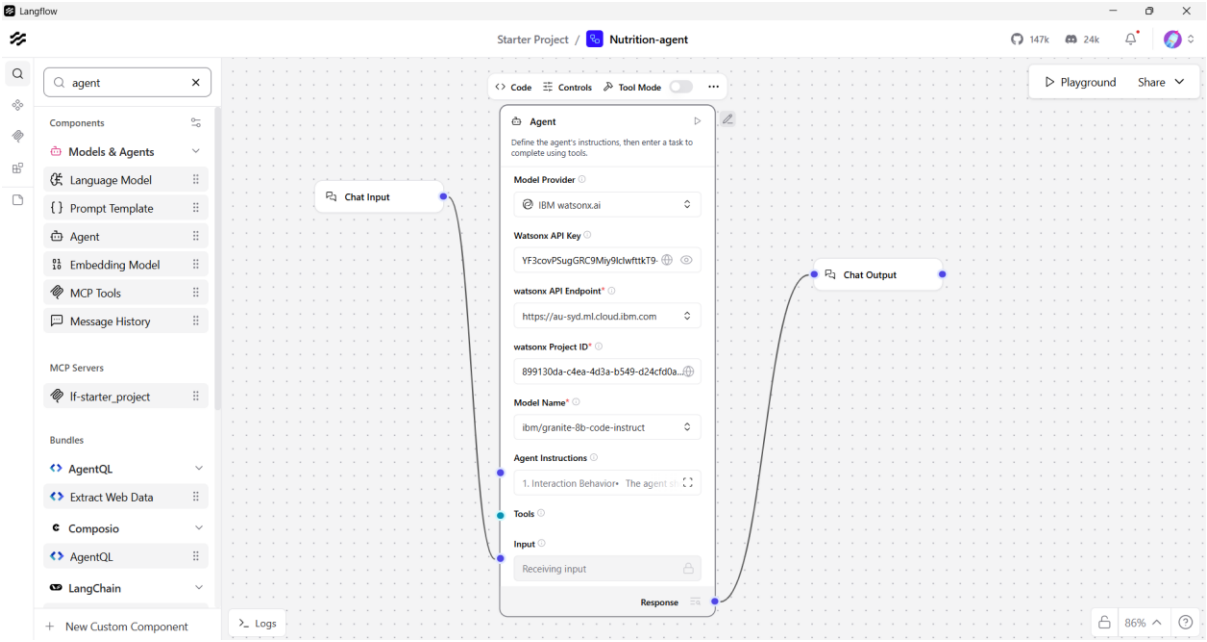
- If input is unclear, ask for clarification.
- If unable to generate a plan, explain the limitation politely.
- Never provide incorrect or misleading information.

9. Engagement Behavior

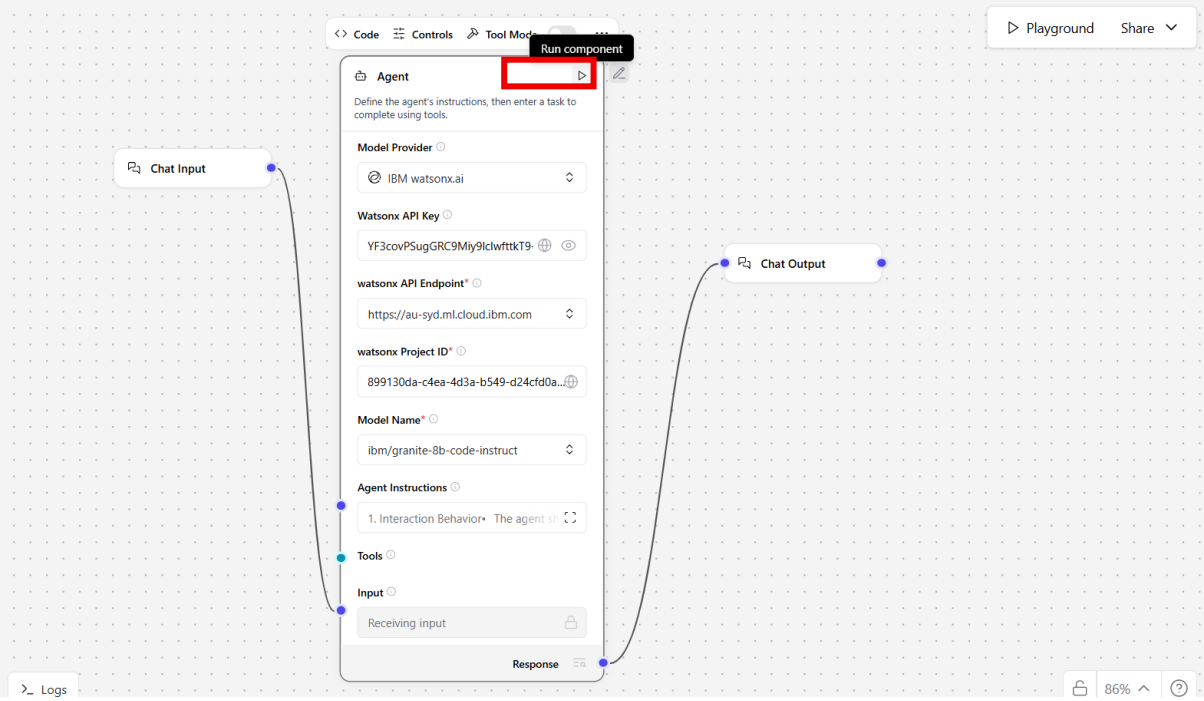
- Encourage user motivation (e.g., "You're doing great!").
- Keep the interaction engaging but professional.
- Offer follow-up help (e.g., weekly plans, grocery lists).



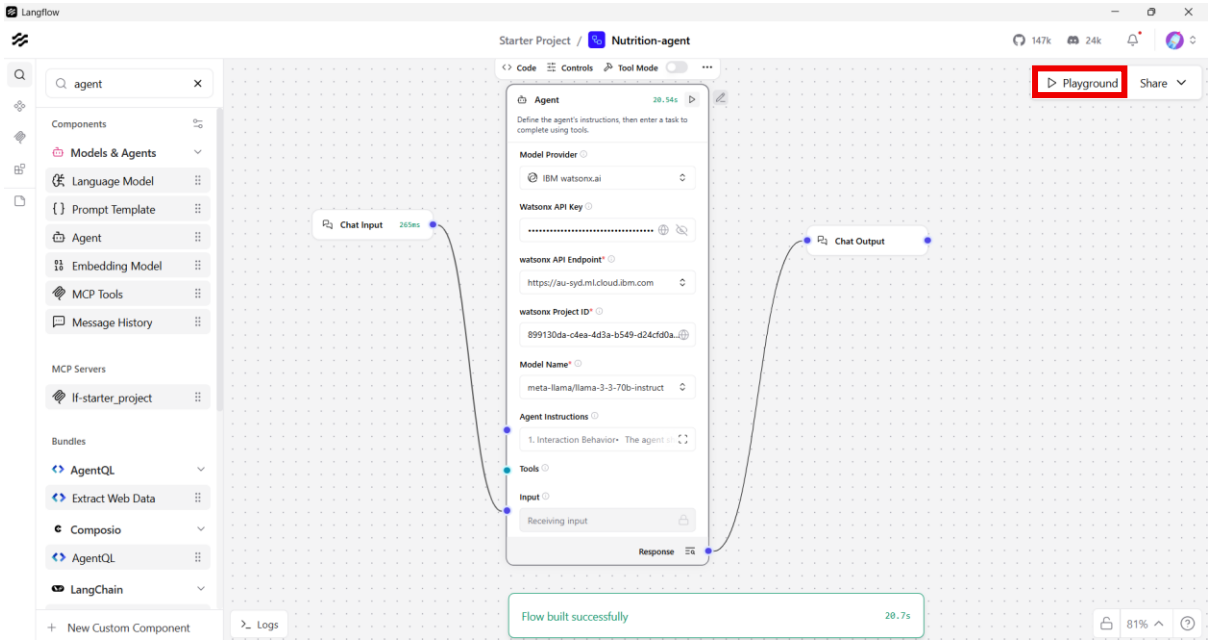
Step-33 select dots and makes connections.



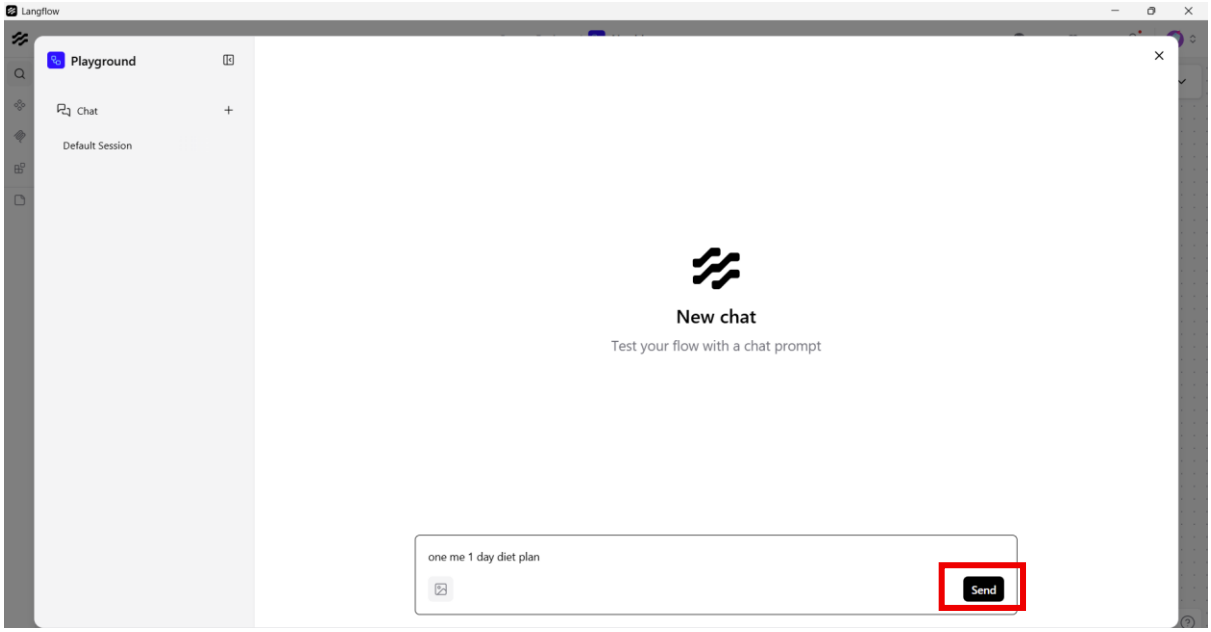
Step-34 Now Click on Run Component Icon.



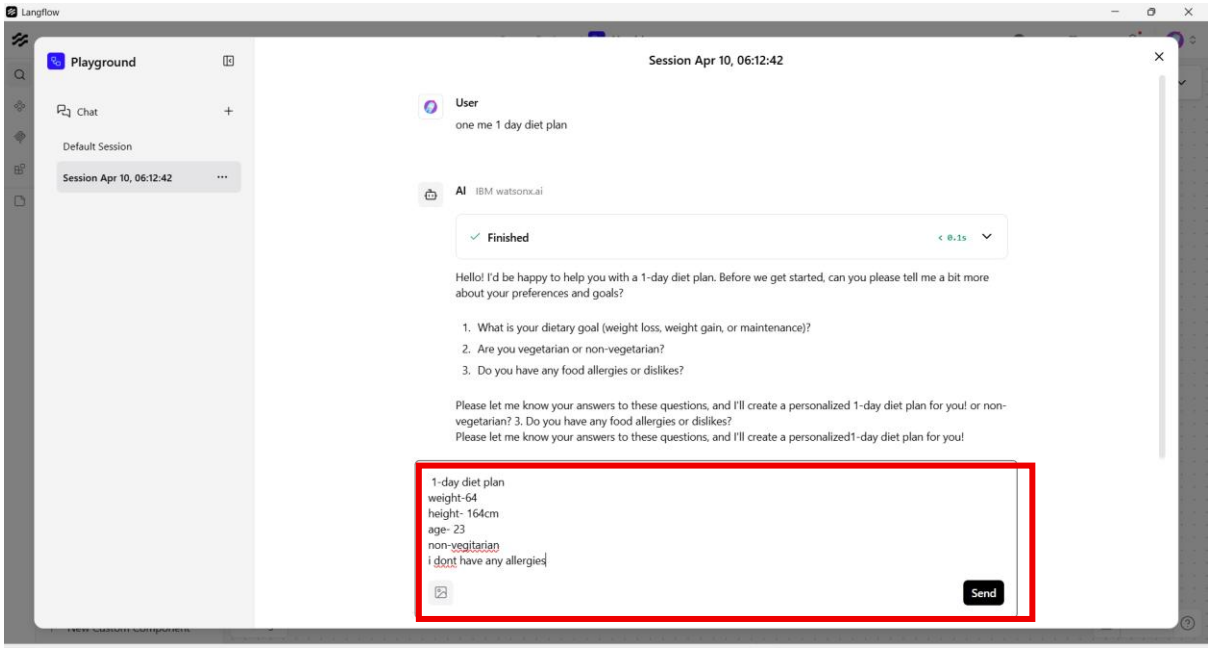
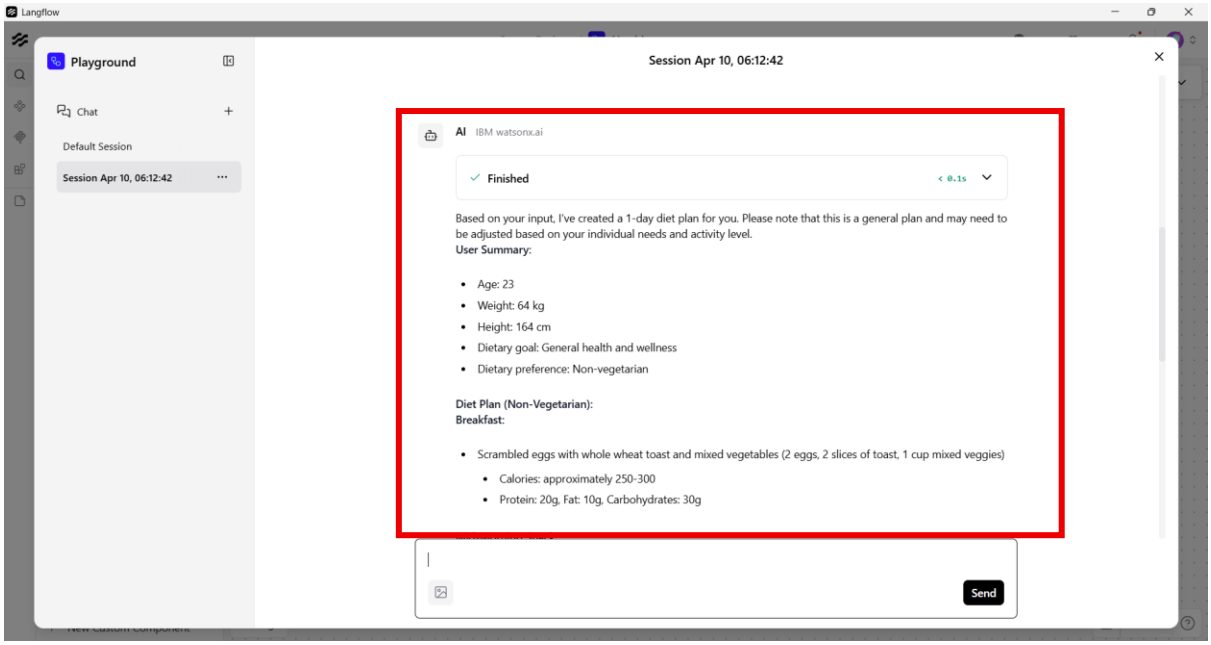
Step-35 once its runs successfully click on playground button.



Step-36 give prompt and test your agent.



Step-34 Now provide needed information



Step-37 Now you can see agent gives me 1 day diet plan.